Exp-8

Part-A

src > JS login.js > default

```javascript
import React, { useState } from "react";

function LoginForm() {
  const [formData, setFormData] = useState({
    username: "",
    email: "",
    password: ""
  });

  const [errors, setErrors] = useState({});
  const [success, setSuccess] = useState("");

  const handleChange = (e) => {
    const { name, value } = e.target;

    setFormData({
      ...formData,
      [name]: value
    });
  };

  const validate = () => {
    let newErrors = {};

    if (!formData.username.trim()) {
      newErrors.username = "Username is required";
    }

    if (!formData.email) {
      newErrors.email = "Email is required";
    } else if (!formData.email.includes("@")) {
      newErrors.email = "Invalid email format";
    }

    if (!formData.password) {
      newErrors.password = "Password is required";
    } else if (formData.password.length < 6) {
      newErrors.password = "Password must be at least 6 characters";
```

```
    }

    return newErrors;
  };

  const handleSubmit = (e) => {
    e.preventDefault();

    const validationErrors = validate();

    if (Object.keys(validationErrors).length > 0) {
      setErrors(validationErrors);
      setSuccess("");
    } else {
      setErrors({});
      setSuccess("Login Successful!");
      setFormData({
        username: "",
        email: "",
        password: ""
      });
    }
  };

  return (
    <div>
      <h2>Login Form</h2>

      <form onSubmit={handleSubmit}>
        <div>
          <input
            type="text"
            name="username"
            placeholder="Username"
            value={formData.username}
            onChange={handleChange}
```

```
        />
        <p style={{ color: "red" }}>{errors.username}</p>
      </div>

      <div>
        <input
          type="t
          name="e
          placeho
          value={formData.email}
          onChange={handleChange}
        />
        <p style={{ color: "red" }}>{errors.email}</p>
      </div>

      <div>
        <input
          type="password"
          name="password"
          placeholder="Password"
          value={formData.password}
          onChange={handleChange}
        />
        <p style={{ color: "red" }}>{errors.password}</p>
      </div>

      <button type="submit">Submit</button>
    </form>

    {success && <p style={{ color: "green" }}>{success}</p>}

    <h3>Entered Data:</h3>
    <p>Username: {formData.username}</p>
    <p>Email: {formData.email}</p>
  </div>
);
}
```

```
const formData: {
    username: string;
    email: string;
    password: string;
}
```

Login Form

Pj44

pksj.4806@gmail.com

•••••••

Submit

Login Successful!

Entered Data:

Username: Pj44

Email: pksj.4806@gmail.com

Part-B

```react/counter.jsx
import React, { useState } from "react";

function Counter() {
  const [count, setCount] = useState(0);
  const [step, setStep] = useState(1);

  const handleStepChange = (e) => {
    const value = Number(e.target.value);
    setStep(isNaN(value) ? 0 : value);
  };

  const increase = () => {
    setCount(count + step);
  };

  const decrease = () => {
    setCount(count - step);
  };

  const reset = () => {
    setCount(0);
  };

  return (
    <div>
      <h2>Counter App</h2>

      <input
        type="number"
        value={step}
        onChange={handleStepChange}
        placeholder="Enter step value"
      />

      <h3>Count: {count}</h3>

      <button onClick={increase}>Increase</button>
      <button onClick={decrease}>Decrease</button>
      <button onClick={reset}>Reset</button>
```

```
      </div>
    );
  }

  export default Counter;
```

Counter App

1

Count: 1

Increase Decrease Reset